

Matrices: Engineering Mathematics Study Notes

3. Eigenvalues and Eigenvectors

High-Level Overview

The concept of eigenvalues and eigenvectors is crucial for engineers and scientists, finding applications in vibration analysis, stability theory, systems of differential equations, and principal component analysis.

Geometrically, for a square matrix A , an **eigenvector** is a non-zero vector X whose direction remains unchanged (or is strictly reversed) when multiplied by A . Mathematically, multiplying A by X merely scales the vector by a scalar value λ , known as the **eigenvalue**.

Key Definitions & Fundamental Equation

- **Eigenvalue Problem:** Given a square matrix A , finding scalars λ (real or complex) and non-zero vectors X such that:

$$AX = \lambda X \iff (A - \lambda I)X = 0$$

where I is the n^{th} order identity matrix.

- **Characteristic Matrix & Equation:**
 - The matrix $(A - \lambda I)$ is the **characteristic matrix**.
 - The determinant equation $|A - \lambda I| = 0$ is the **characteristic equation**.
 - Expanding the determinant yields a polynomial of degree n :

$$(-1)^n \lambda^n + k_1 \lambda^{n-1} + k_2 \lambda^{n-2} + \dots + k_n = 0$$

- **Alternative Names for Eigenvalues:** Characteristic roots, latent roots, or proper values.

Important Properties of Eigenvalues and Eigenvectors

1. Any square matrix A and its transpose A^T share the **exact same eigenvalues**.
2. The eigenvalues of a **triangular matrix** (upper or lower) are simply its **diagonal elements**.
3. The eigenvalues of an **idempotent matrix** ($A^2 = A$) are either **0 or 1**.
4. The **sum of the eigenvalues** equals the **trace** of the matrix (sum of the principal diagonal elements).
5. The **product of the eigenvalues** equals the **determinant** of the matrix ($|A|$).
6. If λ is an eigenvalue of A , then $\frac{1}{\lambda}$ is the eigenvalue of A^{-1} .
7. If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A , then A^m has eigenvalues $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$ for any positive integer m .
8. The eigenvalues of a **real symmetric matrix** are always **real numbers**.
9. The eigenvalues of a **skew-symmetric matrix** are either **purely imaginary or zero**.

Cayley-Hamilton Theorem

Statement: Every square matrix satisfies its own characteristic equation.

If the characteristic equation of an n^{th} order matrix A is given by:

$$(-1)^n \lambda^n + k_1 \lambda^{n-1} + k_2 \lambda^{n-2} + \dots + k_n = 0$$

Replacing λ with matrix powers yields the matrix equation:

$$(-1)^n A^n + k_1 A^{n-1} + k_2 A^{n-2} + \cdots + k_n I = 0$$

- **Application to Find Inverse:** Multiplying the characteristic matrix polynomial by A^{-1} provides a practical, efficient method to compute large matrix inverses using only powers of A up to $n - 1$.

Worked Examples

Example 1: Finding Eigenvalues and Eigenvectors of a 2×2 Matrix

Find the eigenvalues and eigenvectors of matrix $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$.

Step 1: Find the Characteristic Equation

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 2 \\ 4 & 3 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(3 - \lambda) - (2)(4) = 0$$

$$\lambda^2 - 4\lambda - 5 = 0 \implies (\lambda - 5)(\lambda + 1) = 0$$

- **Eigenvalues:** $\lambda = 5, -1$

Step 2: Eigenvector for $\lambda = -1$

Set up $(A + I)X = 0$:

$$\begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Using row reduction, this simplifies to $x + y = 0 \implies y = -x$.

- Letting $x = k$, the eigenvector is $X_1 = k \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Thus, the vector is $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Step 3: Eigenvector for $\lambda = 5$

Set up $(A - 5I)X = 0$:

$$\begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This simplifies to $-4x + 2y = 0 \implies y = 2x$.

- Letting $x = k$, the eigenvector is $X_2 = k \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Thus, the vector is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Example 2: Short-Cut for 3×3 Characteristic Equations

For a 3×3 matrix A , the characteristic equation can be quickly constructed using the formula:

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$$

- $S_1 = \text{trace}(A)$ (sum of main diagonal elements)
- $S_2 =$ sum of the minors of the leading diagonal elements
- $|A| =$ determinant of A